

2.6 Solving Two-Step, Real-World Problems With Fractions or Decimals

Lesson Introduction

 Benchmark	MA.6.NSO.2.3 Solve multi-step, real-world problems involving any of the four operations with positive multi-digit decimals or positive fractions, including mixed numbers.		 Learning Target	I can solve real-world problems with two steps using decimals or fractions.
 Benchmark Coherence	Building On		Working Toward	
	MA.5.AR.1 Solve problems involving the four operations with whole numbers and fractions.		MA.7.NSO.2.3 Solve real-world problems involving any of the four operations with rational numbers.	
 Essential Questions	- What are the similarities and differences between solving one-step and two-step real-world problems? - What different strategies can be used to solve two-step equations with decimals and fractions?			
 Lesson Narrative	In this lesson, students continue applying their knowledge of operations on rational numbers to solve two-step problems. Then, students collaborate by using feedback to improve on their solutions. Students then use the problem-solving process and skills from previous lessons, including converting rational numbers and writing numeric expressions for real-world problems to solve two-step real-world problems involving decimals and fractions.			
 Component Summary	2.6.1 Warm-Up (~5 min.)	Would You Rather Activity (SE p. 65)	2.6.3 Collaboration (10-15 min.)	Error analysis, correction, and growth mindset message on two-step, real-world problems (SE p. 66)
	2.6.2 Guided Practice (10-15 min)	Practice solving two-step equations through scaffolded problems (SE p. 65)	2.6.4 Your Turn! (15-20 min.)	Practice solving two-step, real-world problems (optional Gallery Walk opportunity) (SE p. 67)
	2.6.5 Wrap-Up (~5 min)	Confidence scale for solving real-world problems with decimals and fractions (SE p. 68)		
 Resources	Glossary		Materials	
	<u>Key Terms:</u> N/A <u>Additional Terms:</u> N/A		(Optional) Chart paper (Optional) Markers	
			Additional Preparation	
			N/A	

 Teacher Tips	Common Misconceptions • Students may create inaccurate numeric expressions. • Students may make computational errors when performing operations on multi-digit decimal and fraction problems. • Students may incorrectly round their answer for money problems.	Things to Consider • To foster flexible thinking and problem-solving strategies, ask students to draw a visual model first and then to create a numeric expression from the visual model. • Students may need to round to the nearest hundredth for money problems so that solutions make sense. • Students have additional practice for fluency with decimals and fractions throughout the rest of the unit. Leverage the 2.6.5 Wrap-Up in this lesson as an opportunity to facilitate growth mindset and self-reflection. • Encourage students to use problem-solving strategies and not processes that involve using only keywords or shortcuts.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Number Talk 2.6.1 Warm-Up offers a unique opportunity for students to engage in meaningful mathematical discourse in a friendly way – with Skittles. Consider what this entry point (even with real Skittles) could do for activating students’ engagement with decimals prior to the lesson.	MA.K12.MTR.4.1: Discussion and Reflection In 2.6.3 Collaboration, students analyze the mathematical thinking of another student. Consider the clarification language in MA.K12.MTR.4.1 and how that can help frame important mathematical dialogue as students discuss their thinking, sequence and present their findings, and develop methods of comparison and justification when analyzing work.
 Differentiate Instruction	For 2.6.4 Your Turn!, questions 1 and 3 use fractions while questions 2 and 4 demonstrate decimal fluency. Consider how to best engage your students in sequencing which questions they work on and in what order. Students could complete one of each before comparing with a peer or students could engage in choosing which they feel least or most confident in first. For 2.6.4 Your Turn!, encourage visual learners to draw models prior to creating numerical expressions. For students who are struggling with the computations involving fraction or decimal parts, consider replacing the fractions and decimals with whole numbers, allowing them to focus on understanding the problem-solving process rather than the computation.	
Lesson Notes:		

2.6.1 Warm-Up: Would You Rather?

Component Overview: Students compare two different deals for Skittles to determine the best purchase. Consider using Think Pair Share literacy routines to give students to work individually then hear others' strategies or ideas in a class discussion.



2.6.1 Warm-Up: Would You Rather?

Would you rather buy 8 Skittles for 25 cents or 62 Skittles for \$1.49?



1. Explain your choice.
Answers will vary.
2. Would your answer change if you could choose what flavor the 8 Skittles were but you could not choose the flavors from the bag?
Answers will vary.



Students should be applying skills of estimation in ranges of fractional or decimal quantities. Consider leveraging some of their responses from this activity later in the lesson when grappling with similar fraction and decimal quantities.

- “I figured there are 8 times as many skittles in the second offer, but the price is less than 8 times the 25 cents.” (ratios)
- “I used division to figure out how much 1 skittle cost in each, with the second offer being less cost per skittle.” (unit rate)

2.6.2 Guided Practice: Two-Step Rational Problems

Component Overview: Students practice solving two-step equations through scaffolded problems. Consider having students work in small groups or partnerships to engage in math discussions and check work for accuracy for question 1, then work individually to complete question 2. After question 1, engage in class discussion on strategies used and compare strategies for effectiveness.



2.6.2 Guided Practice: Two-Step Rational Problems

1. Sophie grew her hair out for a long period of time. Her hair is $2\frac{3}{5}$ feet long. She donated her hair by cutting $\frac{5}{6}$ of a foot off. Sophie's hair then grew $1\frac{2}{3}$ feet.
 - a. What operations are needed to determine the length of Sophie's hair?
Addition and Subtraction
 - b. What values are necessary to determine the length of Sophie's hair?
 $2\frac{3}{5}, \frac{5}{6}, 1\frac{2}{3}$
 - c. Write the numeric expression that can be used to determine the length of Sophie's hair.
 $2\frac{3}{5} - \frac{5}{6} + 1\frac{2}{3}$
 - d. Determine the length of Sophie's hair.
 $3\frac{13}{30}$ ft
 - e. Determine if the length of Sophie's hair makes sense. Support your answer.
Answers will vary. No, the length of Sophie's hair does not make sense. It seems too long.
2. Tiffany has a balance of \$167.32 in her checking account. She deposits \$58.16 in her checking account and then writes a check for \$97.27.
 - a. What operations are needed to determine the amount of money in Tiffany's bank account?
Addition and Subtraction
 - b. What values are necessary to determine the amount of money in Tiffany's bank account?
 $\$167.32, \$58.16, \$97.27$
 - c. Write the numeric expression that can be used to determine the amount of money in Tiffany's bank account.
 $\$167.32 + \$58.16 - \$97.27$
 - d. Determine the amount of money Tiffany has left in her checking account.
 $\$128.21$



Notice the inherently scaffolded prompts in this Guided Practice. While monitoring students' practice, consider asking probing questions as informal observational data for the remainder of the lesson for the following questions.

- Question 1a: How do you know? What will that look like for letter C?
- Question 1b: What do you notice about the fractions? What does that make you think about when adding and subtracting those values?
- Question 1d: Does that mean your answer is wrong? Why or why not?



Consider providing grid or graph paper for students to organize their work when calculating with decimals.



Though these problems involve fluency with decimal and fraction operations, position students to focus on the lesson's learning targets as they complete these problems individually. To facilitate this focus, ask questions like:

- How do you know what operation is required?
- How do you know how many steps are involved in finding the final answer?
- How can you check your answer for feasibility within the given context?

2.6.3 Collaboration: Error Analysis

Component Overview: Students complete an Error Analysis task where they analyze and explain the errors, then provide a brief note explaining the mistake and how to learn from it. Group students in partnerships as they evaluate the work, then complete the problem correctly and provide reasons for their solution. Consider combining partnerships for the feedback in question 1c so students share their thinking with others.



2.6.3 Collaboration: Error Analysis

1. Stefanie solved the following word problem. Her work is shown.

- a. Evaluate her work and determine where she made mistakes.

Problem	Stefanie's Work
Janelle has $2\frac{1}{4}$ pans of brownies. Each pan made 8 brownies. She gave $\frac{2}{3}$ of the brownies to her friends. How many brownies did Janelle give away?	Step 1: $2\frac{1}{4} \times 8 = \frac{9}{4} \times 8 = \frac{18}{4}$ Step 2: $\frac{18}{4} \div \frac{2}{3} = \frac{18}{4} \times \frac{3}{2} = 6\frac{3}{4}$ Answer: She gave away $6\frac{3}{4}$ brownies.

- b. Help Stefanie understand and learn from her mistakes. Write a brief note to Stefanie explaining what her mistakes were and how she could learn from those mistakes.

Answers may vary.

Means
I
Start
To
Acquire
Knowledge
Experience
Skills

Possible answer:

When finding a fraction of an amount, use multiplication. When multiplying a fraction by a whole number, change the whole number to a fraction with denominator of 1.

Solution: $(2\frac{1}{4} \times 8) \times \frac{2}{3} = 12$

- c. Read your partner's note to Stefanie and offer suggestions for how to improve their note.



MA.K12.MTR.4.1: Discussion and Reflection
Stefanie makes a computational error in Step 1 when determining how many total brownies are available to give away. Encourage students not only to identify the error, but also to highlight the correct use of multiplication to determine the number of brownies, as well as the opportunity to assess that $\frac{18}{4}$ brownies was not reasonable given the context.



Consider the following probing questions as students collaborate on identifying the error:

- Why is $2\frac{1}{4}$ multiplied by 8? Does the answer $\frac{18}{4}$ make sense? Why or why not?
- Why is that product then divided by $\frac{2}{3}$? Why not addition or multiplication?
- Does the answer seem reasonable?

2.6.4 Your Turn!

Component Overview: Consider using a Gallery Walk for this activity. Provide small groups of 2 to 4 with chart paper and markers. Have students solve all four problems on their chart paper. After about 15 minutes of small group work time, have students display their chart paper on a visible wall/desk, then have the class walk around to view other groups' posters for about 5 minutes. Ask students to visit at least three posters and confirm the work on the poster with their own answer and compare strategies used.



2.6.4 Your Turn!

1. Martha fills her ice cream bowl with $\frac{3}{4}$ cup of ice cream, $\frac{1}{8}$ cup of candy toppings, and the rest with fruit toppings. If the bowl contains a total of $1\frac{1}{2}$ cups, how much of the bowl is made up of fruit toppings?

$\frac{5}{8}$ cups of fruit topping

2. Jasmine has a ribbon that is 12.6 meters long. Sharon has a ribbon that is 3.8 meters shorter. What is the total length of the two ribbons?

21.4 meters

3. Before soccer practice, Hernando puts $4\frac{1}{8}$ cups of sports drink in a water bottle with $1\frac{2}{3}$ cups of ice. After a hot afternoon of practice, he drinks $2\frac{1}{5}$ cups of the sports drink and melted ice. How many cups of this mixture are left in the water bottle?

$3\frac{71}{120}$ cups

4. A bag of chocolates weighs 1.384 pounds. If Luis gives each of his 6 friends 0.23 pounds of chocolate, how many pounds of chocolate does Luis have left?

0.004 pounds



Encourage students to draw visual models prior to attempting to solve the problem. This may help students make connections from a pictorial representation to the numeric expression and operation needed to solve.



To encourage mathematical reasoning, ask questions like "Do your answers make sense? What methods can you use to check your answer?" (MA.K12.MTR.7.1)



As students work on answering the questions, circulate around the room and listen for how students are making sense of them. To develop strategies for understanding problems prior to solving, ask questions like:

- How do you know what you are being asked to find?
- Which step/operation do you need to use first? Which operation/step do you use second? How do you know?
- What strategy can you use? Does drawing a model make sense for this problem?
- Is your answer reasonable? How do you know?

2.6.5 Wrap-Up: Reflection

Component Overview: Students rate confidence with solving real-world, two-step word problems. Student responses and class observations collected from the lesson combined with this reflection to assess student understanding for small group intervention.



2.6.5 Wrap-Up: Reflection

Place an **X** on the line to indicate your understanding of each of today's learning targets. Then, provide a brief explanation or evidence to support your self-assessment.

1. I can solve real-world problems with two steps using decimals.

Answers will vary.

I need help. I'm getting there. I understand
I can't get started. and have evidence.



2. I can solve real-world problems with two steps using fractions.

Answers will vary.

I need help. I'm getting there. I understand
I can't get started. and have evidence.



Consider scaffolded prompts for students to more accurately reflect on their progress.

- *I need help: "I don't know where to start" or "Decimals and fractions are difficult for me"*
- *I'm getting there: "I get some problems right and some wrong" or "I know how to solve but sometimes struggle with decimals"*
- *I understand and have evidence: "I could explain how to do this to someone else" or "Provide an example to show evidence"*